# M.I.N.E.R.S. Rover - V1326 Motor Driver & 12V Battery Wiring Guide

Hardware Configuration: V1326 / L298N Dual H-Bridge, Dedicated 12V Battery, 12V DC Gear Motors, Arduino Mega 2560

**CRITICAL SAFETY RULE: COMMON GROUND REQUIREMENT**

The Dedicated 12V Battery Negative (-) terminal MUST be connected to BOTH the V1326 GND terminal AND an Arduino Mega GND pin. Without this common ground reference, the Arduino control signals will float, causing un-predictable motor behavior or failure to spin.

## 1. Power & Battery Connections (Dedicated 12V Motor Battery)

| V1326 Driver Terminal | Connection Source / Destination | Cable / Notes |
|---|---|---|
| 12V / VCC (+12V Input) | 12V Battery POSITIVE (+) | Thick Wire (Power (+) for 12V Motors) |
| GND (Power Ground) | 12V Battery NEGATIVE (-) | Thick Wire (Power Ground (-)) |
| GND (Signal Ground) | Arduino Mega GND Pin | MUST connect to Arduino GND for Common Ground |
| 5V Terminal | LEAVE DISCONNECTED | Arduino powered via USB/separate 5V supply |

## 2. Arduino Mega 2560 Control Signals (mega2560_rover_master.ino)

| V1326 Driver Pin | Arduino Mega Pin | Function in Code | Default Jumper State |
|---|---|---|---|
| ENA (Left Speed) | Pin D2 (PWM) | Left Motors Speed PWM (0-255) | REMOVE Jumper Cap |
| IN1 (Left Dir A) | Pin D30 | Left Motors Direction 1 | Male Header Pin |
| IN2 (Left Dir B) | Pin D31 | Left Motors Direction 2 | Male Header Pin |
| IN3 (Right Dir A) | Pin D32 | Right Motors Direction 1 | Male Header Pin |
| IN4 (Right Dir B) | Pin D33 | Right Motors Direction 2 | Male Header Pin |
| ENB (Right Speed) | Pin D3 (PWM) | Right Motors Speed PWM (0-255) | REMOVE Jumper Cap |

## 3. 12V DC Motor Outputs (4WD Parallel Skid-Steer)

| V1326 Screw Terminal | Connected Motors | Wiring Configuration |
|---|---|---|
| OUT1 | Front-Left Motor (+) & Rear-Left Motor (+) | Left Motors Parallel (+) |
| OUT2 | Front-Left Motor (-) & Rear-Left Motor (-) | Left Motors Parallel (-) |
| OUT3 | Front-Right Motor (+) & Rear-Right Motor (+) | Right Motors Parallel (+) |
| OUT4 | Front-Right Motor (-) & Rear-Right Motor (-) | Right Motors Parallel (-) |

## Quick Testing Tip: Direction Check

If any motor rotates backward when driving FORWARD ('W' or 'CMD:F'), simply swap its (+) and (-) wires at the OUT1/OUT2 or OUT3/OUT4 screw terminal block.